

The human factor in IT security - social engineering, local machine (PC) security, and data destruction

1. The human factor in IT security

Definition:

Humans are often the **weakest link** in the cybersecurity chain, making mistakes or falling victim to manipulation.

Common human-related risks:

- Clicking on malicious links
- Using weak passwords
- Leaving devices unattended
- Ignoring security warnings

Key idea: Even the best technology fails if the user isn't educated or vigilant.

2. Social engineering

Definition:

The use of deception to manipulate individuals into revealing confidential information or performing actions that compromise security.

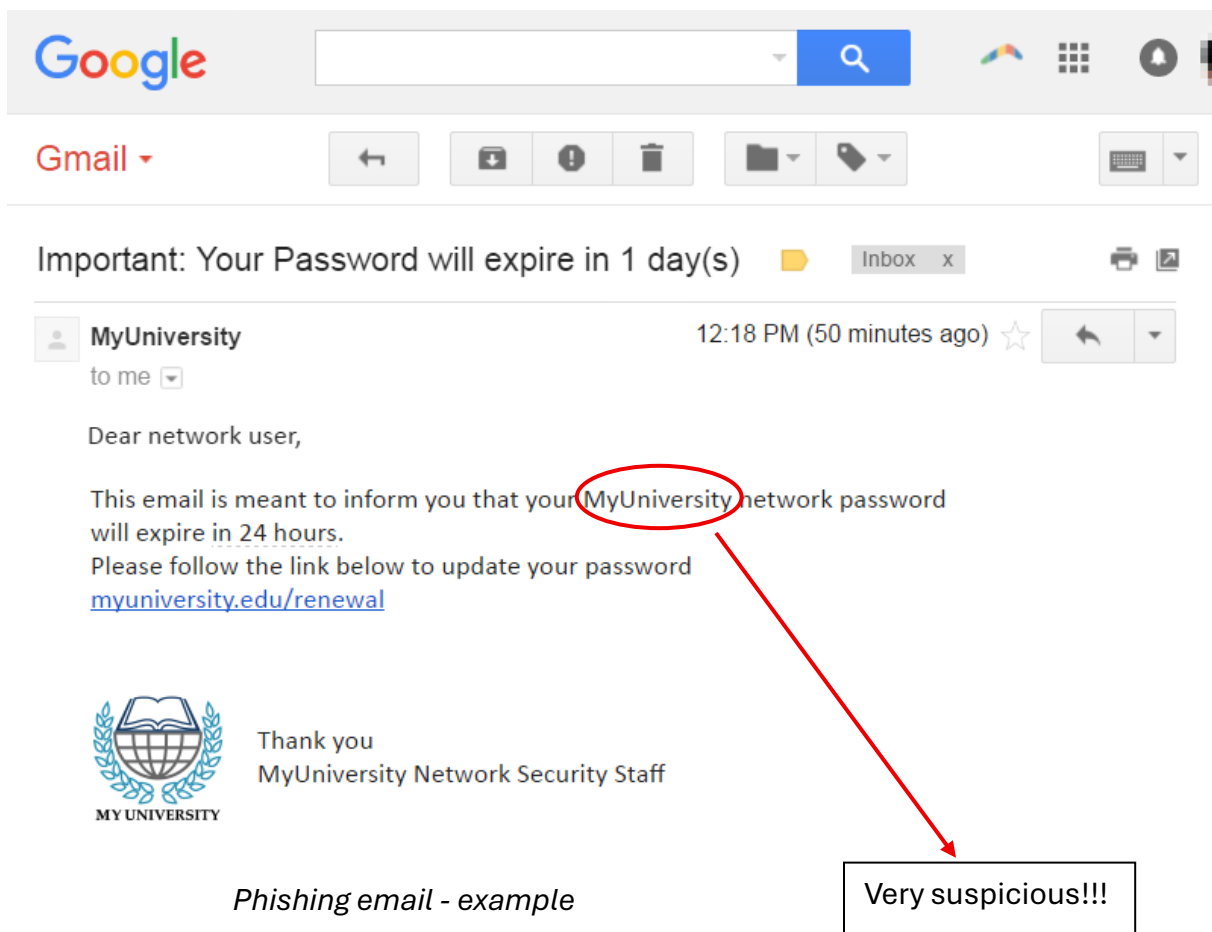
Common techniques:

Technique	Description	Example
Phishing	Fake emails or websites to steal credentials	Email pretending to be from a bank
Pretexting	Inventing a scenario to gain trust	"I'm from IT support, I need your password"

Technique	Description	Example
Baiting	Offering something appealing to trick users	Free USB stick with malware
Tailgating	Physically following someone into a restricted area	Entering a locked room behind an employee
Vishing	Voice phishing via phone	Caller claiming to be tech support

Phishing - real-world example:

The 2011 RSA breach started with a phishing email opened by a single employee. The attacker eventually gained access to classified network credentials.



Phishing email - example

Local security & data destruction

1. Security of local machines (PCs)

Why local protection matters?

Even with strong network security, an unprotected PC is vulnerable to:

- Malware infections
- Unauthorized access
- Physical theft
- Data breaches

Protection options:

Method	Purpose
Antivirus & anti-malware	Detect and remove malicious software
Firewall, intrusion detection / prevention systems	Blocks unauthorized inbound/outbound connections.
OS and software updates (patches)	Fix security vulnerabilities
Strong passwords	Prevent unauthorized login
Disk encryption (e.g. BitLocker)	Protects data if the computer is stolen
BIOS (Basic Input/Output System) / UEFI (Unified Extensible Firmware Interface) passwords	Prevents booting or changing settings without authorization
Auto-lock screens	Reduces risk of unauthorized physical access
USB port blocking	Prevents data theft via removable media

2. Secure data destruction

Why it's important?

Simply deleting a file does not remove it permanently - it can often be recovered with special tools.

Data destruction methods:

Method	Description
Delete	Removes file from file system; recoverable
Formatting	Erases filesystem; still recoverable
Overwriting	Replaces file data with random bits (done multiple times for safety)
Wiping (deletion) tools	Tools like DBAN, Eraser, or built-in OS options
Demagnetization	Disrupts magnetic fields; destroys hard drive data
Physical destruction	Shredding, crushing, or melting the storage device





Tip: SSDs (Solid-State Disk) need different erasure techniques than HDDs (Hard Disk Drives) due to how data is stored.

Best practice: Use both **software wiping** and **physical destruction** when discarding sensitive devices.

Key terms summary

Term	Definition
Human factor	The role of human error or behavior in IT security vulnerabilities
Social engineering	Manipulating people into giving up confidential information
Phishing	Fake messages designed to steal information
Local machine security	Protection measures on individual computers
Data wiping	Overwriting data to prevent recovery
Physical destruction	Physically damaging storage media to destroy data
Firewall	Monitors and controls network traffic
Antivirus	Detects and removes harmful software